PORTFOLIO 2026 · FOUR CASE STUDIES

YASAR C H

where fashion *meets the machine.*



FEATURING GARMENTFIX QMS · APMS · SMART MONITORING · TEDXNIFT JODHPUR

WITH GARMENT DEVELOPMENT · FASHION & EVENT PHOTOGRAPHY

BUILT AT BRANDIX APPAREL INDIA, UNIT 3 · NIFT JODHPUR

FASHION TECHNOLOGY · AUTOMATION · PHOTOGRAPHY

SCAN OR CLICK FOR THE INTERACTIVE CUT



ABOUT

I build systems that make factories *run themselves.*

I am a Bachelor of Fashion Technology student at NIFT Jodhpur, interested in combining fashion, technology, digital development and visual storytelling. I enjoy building digital systems, exploring AI-assisted development, working with technology in manufacturing, creating visual content, and managing creative projects and events.

4
case studies, all built and running

1,486
machines on APMS at Brandix Unit 3

6 h → live
QMS quality reporting delay

<$40
IoT hardware per sewing machine

June – July 2026 · 8 weeks	**Brandix Apparel India Pvt. Ltd.** Apparel Internship · Unit-III, Visakhapatnam
June 2025 · 16-30 June	**Arvind Limited** Textile Manufacturing Training · Ahmedabad
November 2025	**TEDxNIFT Jodhpur** Website Designer & Developer
2026-2027	**Adventure & Photography Club, NIFT Jodhpur** President

EDUCATION

National Institute of Fashion Technology (NIFT), Jodhpur · Bachelor of Fashion Technology (BFT) · Minor: Communication Design

RECOMMENDED BY

Mr. Amit Patil, Principal Design Manager, Microsoft · Mr. Narpath Raman, Mentalist & Mind Reader

INSIDE

Case studies: GarmentFix QMS (p3), APMS (p6), Smart Monitoring (p9), TEDxNIFT Jodhpur (p12) · Garments & photography (p15)

CASE STUDY 01 · BRANDIX APPAREL INDIA · UNIT 3 · VIEW LIVE CASE STUDY ↗

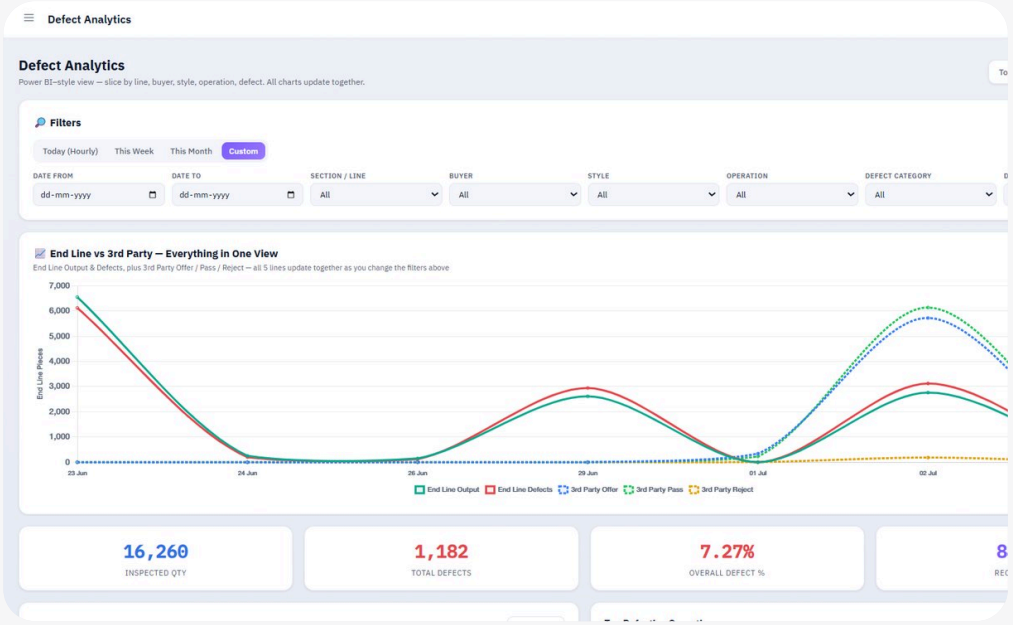
GarmentFix QMS

Real-Time Quality Management System · System Designer & Developer

01 PROBLEM & BRIEF

Quality checks lived on paper. Each shift produced stacks of inspection sheets that someone re-typed for hours, and management saw the numbers six hours late.

BUILT WITH JavaScript · Firebase Realtime Database · QR Code · PWA · Groq AI · AMMS integration



70–80 → 0
paper sheets per shift

3–4 h → 0
manual re-entry per day

6 h → Live
reporting delay

02 AUDIENCE

01
End-line & roving QC
Log every check on a phone at the machine, with no retraining.

02
QC head & managers
See defect rates and risky operators live, not six hours later.

03
Third-party auditors
Their audit stage captured in the same system.

04
Admin
Set styles, operations, the defect library, line rosters and rotation.

02 RESEARCH

What I *found.*

01 — 70–80 paper sheets a shift

End Line, Roving and 3rd Party audits each had their own paper form.

02 — Half a shift re-keying

One designated QC spent half a shift typing sheets in, time meant for inspecting.

03 — A six-hour lag

Totals were calculated by hand and reached the Power BI dashboard about six hours late.

04 — Every form studied field by field

The digital version keeps the factory's own vocabulary and layout, so QC staff needed no retraining.

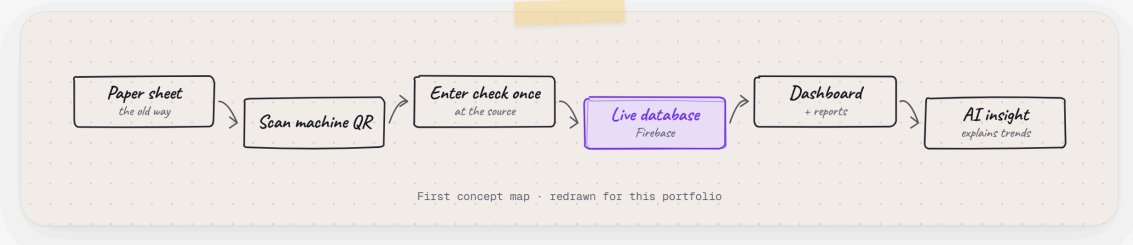
05 — Factory rules, enforced

A defect needs a CPA; visit N+1 cannot start while visit N is missing its action.

Guided by Ms. Bhavani K (HR) and Mr. Vignesh (QC HOD) at Brandix Unit 3.

03 IDEATION & PROCESS

From idea to *system.*



01 Observed the paper process

02 Mapped every form

03 Worker app + admin dashboard

04 Pilot on Line 10

05 Fixed 8 floor issues

06 Ready for factory-wide use

OPTIONS I WEIGHED

CONSIDERED
~~Installing an app or buying scanners~~

CHOSE
One web page that opens in any Android browser

No install, no app store, no new hardware: it runs on the phones already on the floor.

CONSIDERED
~~Flag operators by defect volume~~

CHOSE
Flag by recurrence across visits

Five of one defect in a single visit is usually a machine or a bad bundle; the same defect on visits 1 and 3 points to a person who needs coaching.

CONSIDERED
~~Rank operators from any sample~~

CHOSE
A minimum of 20 pieces before ranking

Nobody appears as worst on the line from a handful of checks.

CONSIDERED
~~Let the AI write the headline numbers~~

CHOSE
Numbers computed in code; AI only explains

"Biggest Risk Right Now" never depends on a model, and the AI is grounded in live figures.

04 EXECUTION & IMPACT

How it *works.*

Inspectors log end-line, roving and third-party checks directly on a device. Defects flow into live analytics, repeat offenders are flagged automatically, and reports are ready the moment the data lands.

01 SETUP

Admin configures

Styles, operations, defect library, line rosters and rotation.

02 INSPECT

End Line

Final check, garment by garment.

Roving

In-line spot checks by operator.

3rd Party

Independent audit stage.

03 LIVE DATA

Firebase Realtime DB

Inspections, defects and rework written the moment they happen.

04 ACT

Red Operator detection

Flags operators repeating the same defect.

Approvals hub

Floor edit requests reviewed by admin.

AMMS ticket

Machine-caused defects go to maintenance.

05 REPORT

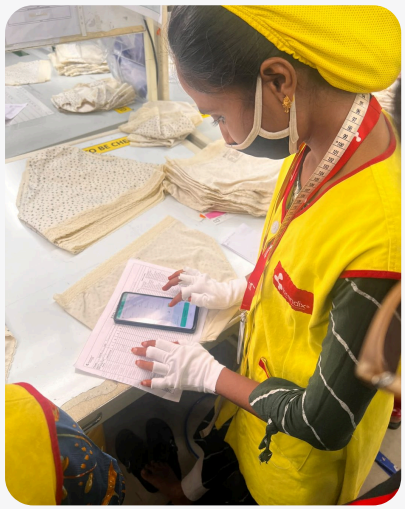
QMS admin dashboard

Defect analytics, operator analysis, module, style and shift reports, corrective actions.

06 INSIGHT

QMS Bot (AI)

Groq-powered analysis that explains trends and answers questions about quality.







Tested on the floor

Piloted on Line 10 during live production, 26 June – 4 July 2026. Eight issues surfaced on the floor; all eight were fixed and re-verified inside the window. Confirmed ready for factory-wide use on 5 July 2026.

CASE STUDY 02 · BRANDIX APPAREL INDIA · UNIT 3 · VIEW LIVE CASE STUDY ↗

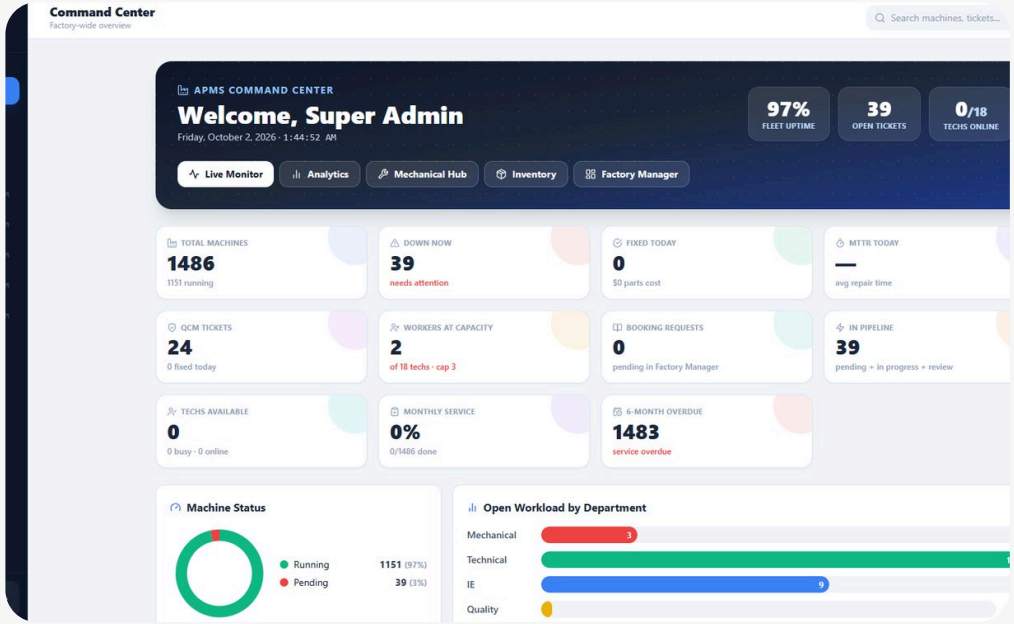
APMS

Automatic Problem Management System · System Designer & Full-Stack Developer

01 PROBLEM & BRIEF

When a sewing machine stops, every minute counts. Breakdowns, maintenance calls and style changeovers were reported by walking, shouting and paper — slow to reach the right engineer and impossible to analyse later.

BUILT WITH React · Tailwind CSS · Firebase Realtime Database · Firebase Cloud Messaging · PWA · QR Code



1,486 machines on the system

32 production modules

4 factory sections

4 engineering departments

02 AUDIENCE

01

Machine operators

Raise a stoppage in one scan, standing at the machine, with no login.

02

Technicians

Mechanical, Technical, IE and Quality each get only their own jobs, pushed to the phone.

03

Stores

Approve and issue spare parts against a live ticket.

04

Admin & factory manager

Open tickets, technicians online, PM due and repair history in one view.

02 RESEARCH

What I *found.*

01 Stoppages were reported on foot

Breakdowns, maintenance calls and style changeovers travelled by walking, shouting and paper.

02 Nothing was timestamped

No record of when a call was raised or fixed, so downtime could not be analysed later.

03 The floor is big

1,486 machines across 32 production modules, 4 sections and 4 engineering departments.

04 Breakdowns are not the only stop

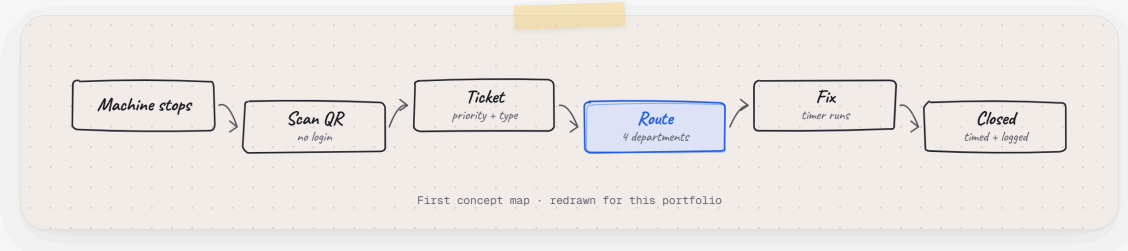
Style changeovers also stop machines, so they had to be tracked alongside breakdowns.

05 Not every mechanic fits every machine

Skill differs by machine type, which became the mechanic skill matrix.

03 IDEATION & PROCESS

From idea to *system.*



01 Map how stoppages were reported

02 Give every machine a QR identity

03 Operator and technician flows

04 Routing, timers and spare parts

05 PM scheduler and skill matrix

06 Linked with QMS defects

OPTIONS I WEIGHED

CONSIDERED

~~Operators log in to report a problem~~

CHOSE

Scan the machine's QR code, no login

At a stopped machine every second counts, and the QR already says which machine it is.

CONSIDERED

~~One shared queue for every engineer~~

CHOSE

Auto-route each ticket to its department

Mechanical, Technical, IE and Quality only see and accept their own work.

CONSIDERED

~~A native app to install on each phone~~

CHOSE

An installable PWA

Runs on the phones already on the floor, updates instantly, nothing to roll out.

CONSIDERED

~~Fix machines only after they fail~~

CHOSE

A preventive maintenance scheduler

Service tickets are raised from the last service date, before a breakdown.

04 EXECUTION & IMPACT

How it *works.*

Operators scan a QR code on the machine and raise the problem in seconds. The right engineering department is pushed a notification instantly, and every event is timestamped from call to fix.

01 REPORT

Scan machine QR

Every machine carries a QR code. The operator scans it on the floor, with no login needed.

02 TICKET

Raise the problem

Breakdown or changeover, with priority and a short description.

03 LIVE DATA

Firebase Realtime DB

Machine status, tickets and every timestamp, synced live to all screens.

04 ROUTING

Mechanical

Technical

IE

Quality

05 FIX

Technician app

Push alert → accept → fix with a running timer → close or escalate.

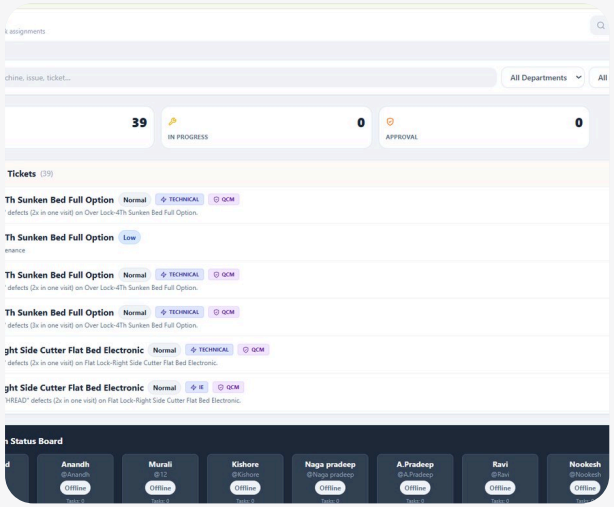
Spare parts

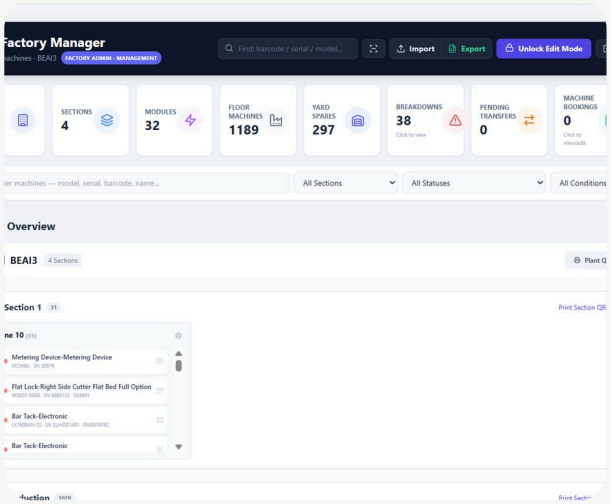
Part requests go to stores for approval and issue.

06 OVERSIGHT

Admin & factory manager

Open tickets, techs online, repair logs, PM due, reports; machines, modules and transfers.





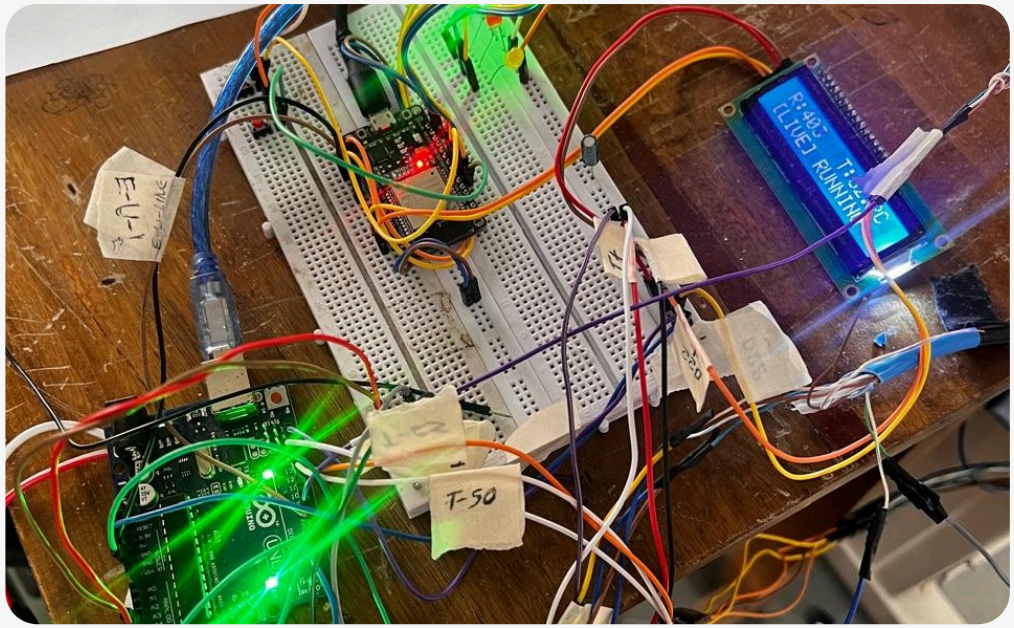
Key features

QR machine identity · Instant push alerts · Changeover tracking · Works like an app · Command Center · Live Monitor.

CASE STUDY 03 · INDUSTRIAL IOT · VIEW LIVE CASE STUDY ↗

GarmentFix Smart Monitoring

Smart Condition Monitoring for Industrial Sewing Machines · System Designer & Developer



01 PROBLEM & BRIEF

Sewing machines run until they visibly break. Bearing and belt wear, overheating motors and snapped needles are found only after the failure, logs live on paper if at all, and one stopped machine can leave 20–30 workers on a line sitting idle.

BUILT WITH Arduino Uno · ESP32 · C++ · CAN bus (SN65HVD230) · Firebase Realtime Database · MPU6050

<$40	2.5 G	75 °C	5 s
hardware per machine	vibration fault threshold	motor temp: automatic power cut	watchdog before SYS_ERROR

02 AUDIENCE

01
Operators
An LCD and andon light at the machine, plus needle-change and e-stop buttons.

02
Mechanics
Vibration and heat warnings before the machine fails.

03
Supervisors
The live state of every machine: running, idle, needle change, emergency, off.

04
Managers
True OEE and PDF shift reports, without anyone filling them in.

02 RESEARCH

What I *found.*

01 Machines run to failure

Bearing and belt wear, overheating motors and snapped needles were found only after the failure.

02 One stop idles a line

A single stopped machine can leave 20–30 workers sitting idle.

03 Logs lived on paper

If they were kept at all, so patterns were invisible.

04 The CAN bus had no documentation

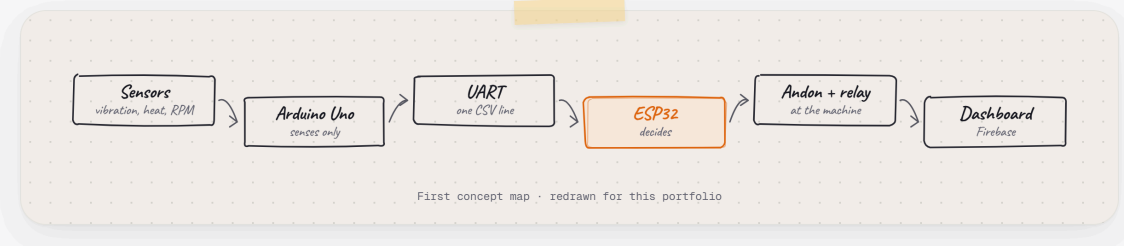
Raw frames were logged until the sleep-packet pattern showed when the machine was really on.

05 It had to be a retrofit

Nothing could touch the machine's control logic, and the hardware had to stay under $40.

03 IDEATION & PROCESS

From idea to *system.*



01 Studied how machines fail

02 Reverse-engineered the CAN bus

03 Breadboard prototype

04 Fitted on a running machine

05 Live dashboard + PDF reports

06 Phase 2 PCB

OPTIONS I WEIGHED

CONSIDERED

~~One microcontroller doing everything~~

CHOSE

Two boards: Uno senses, ESP32 decides

Keeping the Uno off the network keeps its timing perfect, so no RPM pulse is dropped.

CONSIDERED

~~A plastic DS18B20 temperature probe~~

CHOSE

A K-type thermocouple with a MAX6675

The DS18B20 cracked under vibration testing; the industrial part did not.

CONSIDERED

~~Jumper wires on a breadboard~~

CHOSE

One compact PCB (Phase 2)

The vibration GarmentFix measures also shakes the jumper wires loose.

04 EXECUTION & IMPACT

How it *works.*

A retrofit, not a replacement. An Arduino Uno reads every sensor; an ESP32 listens to the machine's own CAN bus, runs a strict safety state machine, drives an andon tower light and LCD at the machine, cuts power on overheat, and logs everything to Firebase for a live dashboard and PDF shift reports.

01 SENSE

Vibration
MPU6050

Motor temp
MAX6675 + K-type

Room
DHT11

Speed
shaft pulses

02 NODE A

Arduino Uno
Reads 5 sensors, does the vibration maths, never touches the network. One CSV line over UART.

03 NODE B

ESP32
Wi-Fi + Firebase, CAN bus listener (machine ECU), safety state machine.

04 AT THE MACHINE

Andon + relay

16×2 LCD

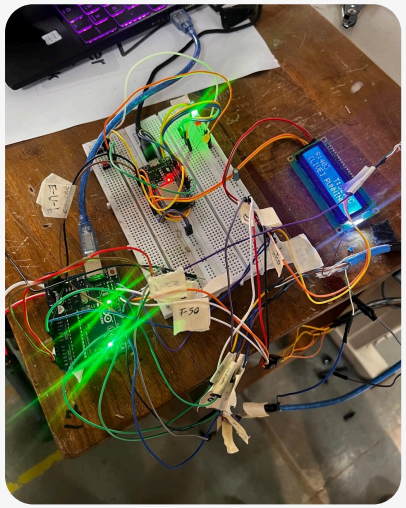
Needle & e-stop buttons

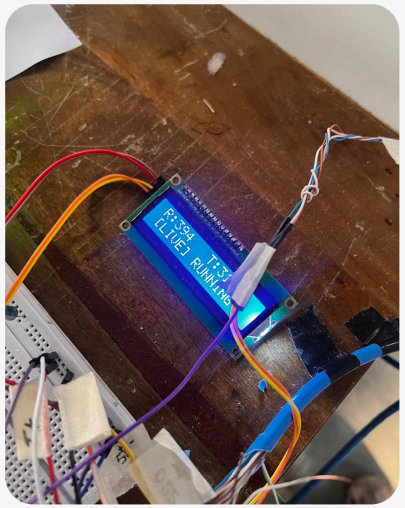
05 CLOUD

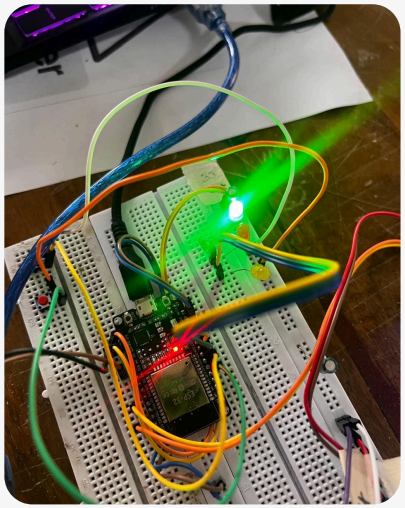
Firebase Realtime DB
History, settings, live shift state and archived shift reports.

06 USE

Live dashboard
OEE, trends and events for 4 roles, plus ISO-style PDF shift reports.







Where it stands

Done · Phase 1: Five sensors reading correctly, CAN bus decoded at 100% accuracy, UART bridge stable, Firebase sync live. In progress · Phase 2: Compact PCB with both MCU footprints, Keyed JST connector per sensor, Onboard 24 V → 5 V / 3.3 V, Conformal coating and enclosure.

CASE STUDY 04 · TEDXNIFT JODHPUR · VIEW LIVE CASE STUDY ↗

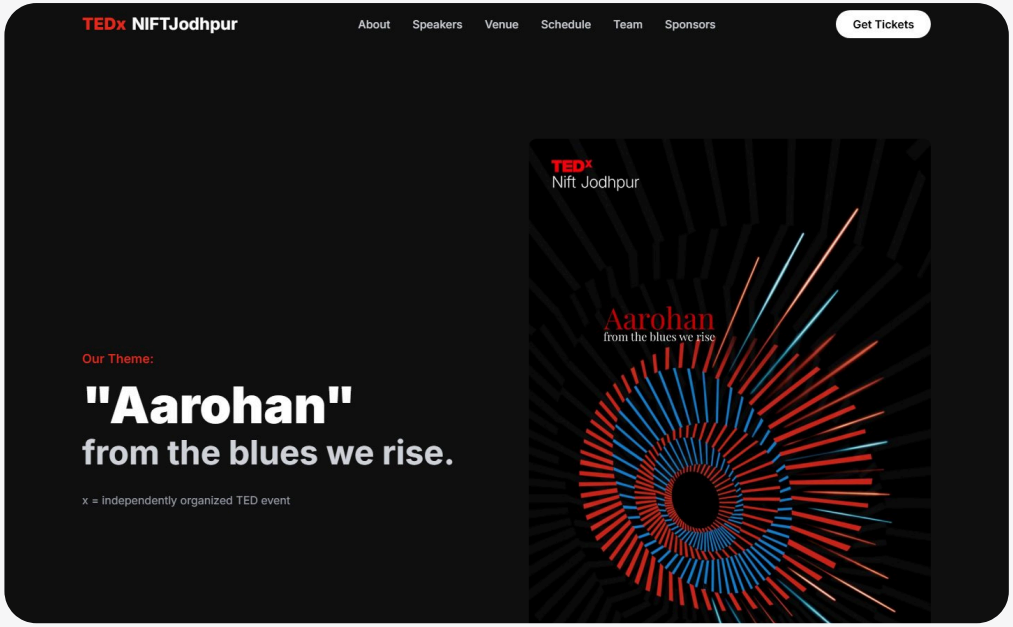
TEDxNIFT Jodhpur

Website & Digital Platform · Website Designer & Developer

01 PROBLEM & BRIEF

An event run by students needed to recruit volunteers across departments, run interviews and seat an audience — without spreadsheets breaking at the last minute.

BUILT WITH HTML · CSS · JavaScript · Firebase · Cloudinary · Vercel



Nov 2025
live for the event

6 in 1
website, applications, interviews, departments, seats, tickets

100%
mobile-first

02 AUDIENCE

01
Attendees
Find the event, choose a seat and carry the ticket on their phone.

02
Volunteer applicants
Apply to a department and follow their interview status.

03
Department leads
Manage only their own team's applicants and volunteers.

04
Organisers
One live view of volunteers, departments and seating.

02 RESEARCH

What I *found.*

01 **Several teams, one intake**

Volunteers applied across departments, each run by its own lead.

02 **Applicants needed answers**

People waiting on interviews needed to see where they stood.

03 **Seating on the day**

The audience needed assigned seats and tickets without last-minute spreadsheet errors.

04 **Everyone is on a phone**

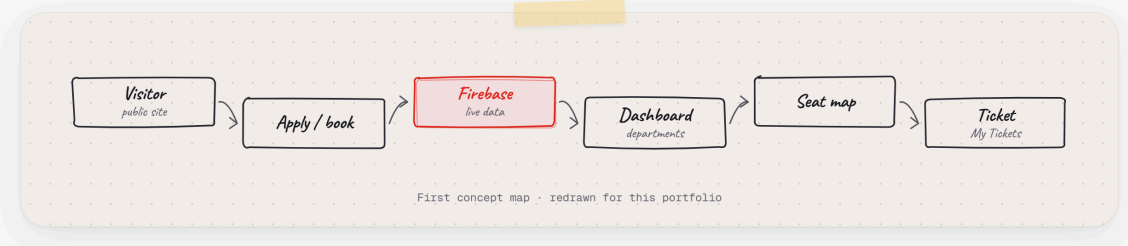
Visitors, volunteers and attendees all arrive on mobile, so every screen is built for it.

05 **Images are heavy**

Event and speaker images had to load fast on mobile data.

03 IDEATION & PROCESS

From idea to *system.*



01 **Brief from the organising team**

02 **Mapped the four user roles**

03 **Public website**

04 **Volunteer system**

05 **Seat map + tickets**

06 **Live for November 2025**

OPTIONS I WEIGHED

CONSIDERED

~~Spreadsheets and separate forms~~

CHOSE

One live platform for every role

Applications, departments, seats and tickets share one source of truth.

CONSIDERED

~~Generic tickets without seats~~

CHOSE

A digital ticket tied to a chosen seat

Attendees pick their own seat on a live map and keep the ticket in My Tickets.

CONSIDERED

~~Images hosted with the site~~

CHOSE

Images served from Cloudinary

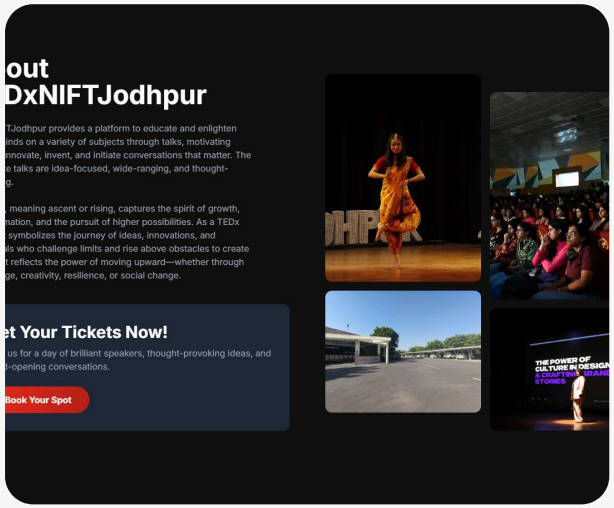
Faster loading on every phone in the audience.

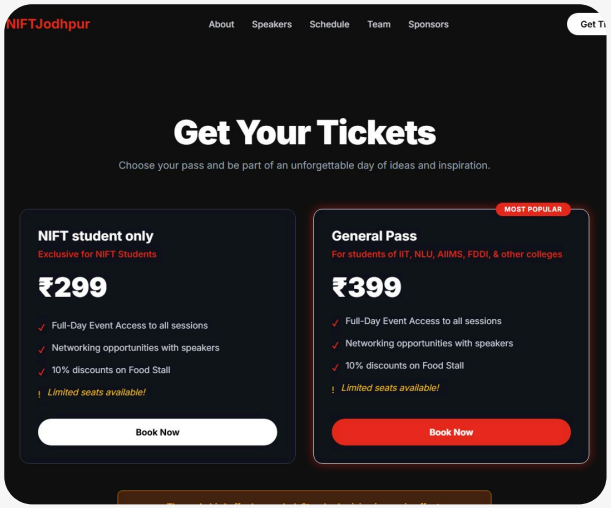
04 EXECUTION & IMPACT

How it *works.*

One platform covers it end to end: volunteers apply and track their interview status, organisers manage departments from a dashboard, and attendees pick a seat and carry their ticket on their phone.

01 VISITORS	02 JOIN	03 DATA	04 ORGANISE	05 ATTEND	06 MEDIA
Public website Event, speakers, venue and schedule.	**Volunteer application** **Interview status**	**Firebase** Applicants, departments, seats and tickets, live.	**Department management** **Volunteer dashboard**	**Seat selection** **Digital ticket** **My Tickets**	**Cloudinary** Images served fast to every phone.





Outcome

Live for the event: Used for TEDxNIFT Jodhpur, November 2025, at apc-movie.vercel.app. Six jobs, one platform: Website, applications, interviews, departments, seating and tickets. Recommended: Mr. Amit Patil, Principal Design Manager at Microsoft, recommends Yasar for his contribution to TEDxNIFT Jodhpur.

GARMENT DEVELOPMENT

Made by *hand.*

Women's Full-Length Dress and Contemporary Crop Top & Skirt: design, pattern making, cutting, construction, detailing, finishing and styling.



PHOTOGRAPHY

Behind the *lens.*

Fashion, runway, brand and event work on a Nikon Z6 II. Clients: The Artsy Harbour and NEWME Jodhpur.



Let's build something *together.*

chyasar2004@gmail.com · yasar-portfolio-three.vercel.app · behance.net/yasarch · linkedin.com/in/yasar-c-h-78131b2b6 · github.com/Yasar011

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